Breaking the Walls:
Open Scale-Across for the AI Era

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October 2026

AI gains compound faster than infrastructure can scale

100+

GPUs were used to trained GPT-2

FloydHub

200K+

GPUs trained xAI Colossus (Grok family)

NVIDIA

2x

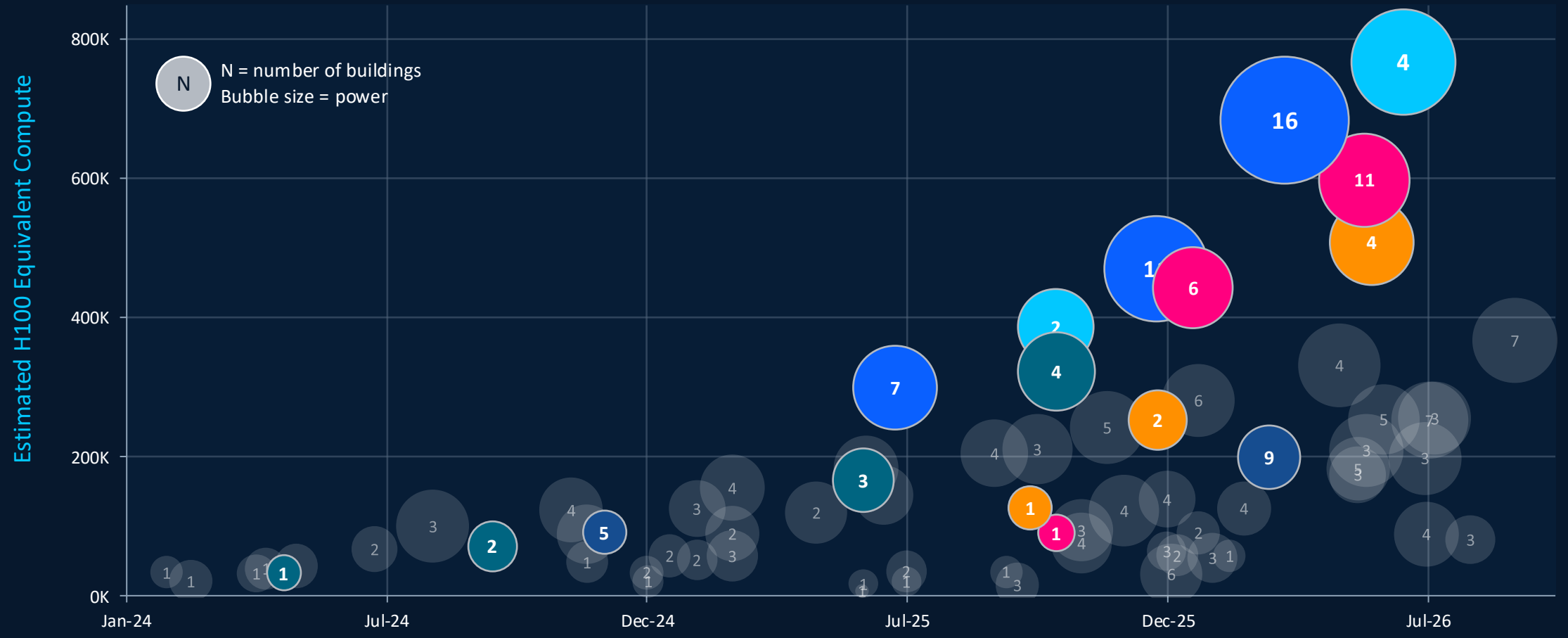
Training compute is doubling roughly every 5 months

Stanford HAI, 2025 AI Index Report

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Scaling AI compute requires power and sites



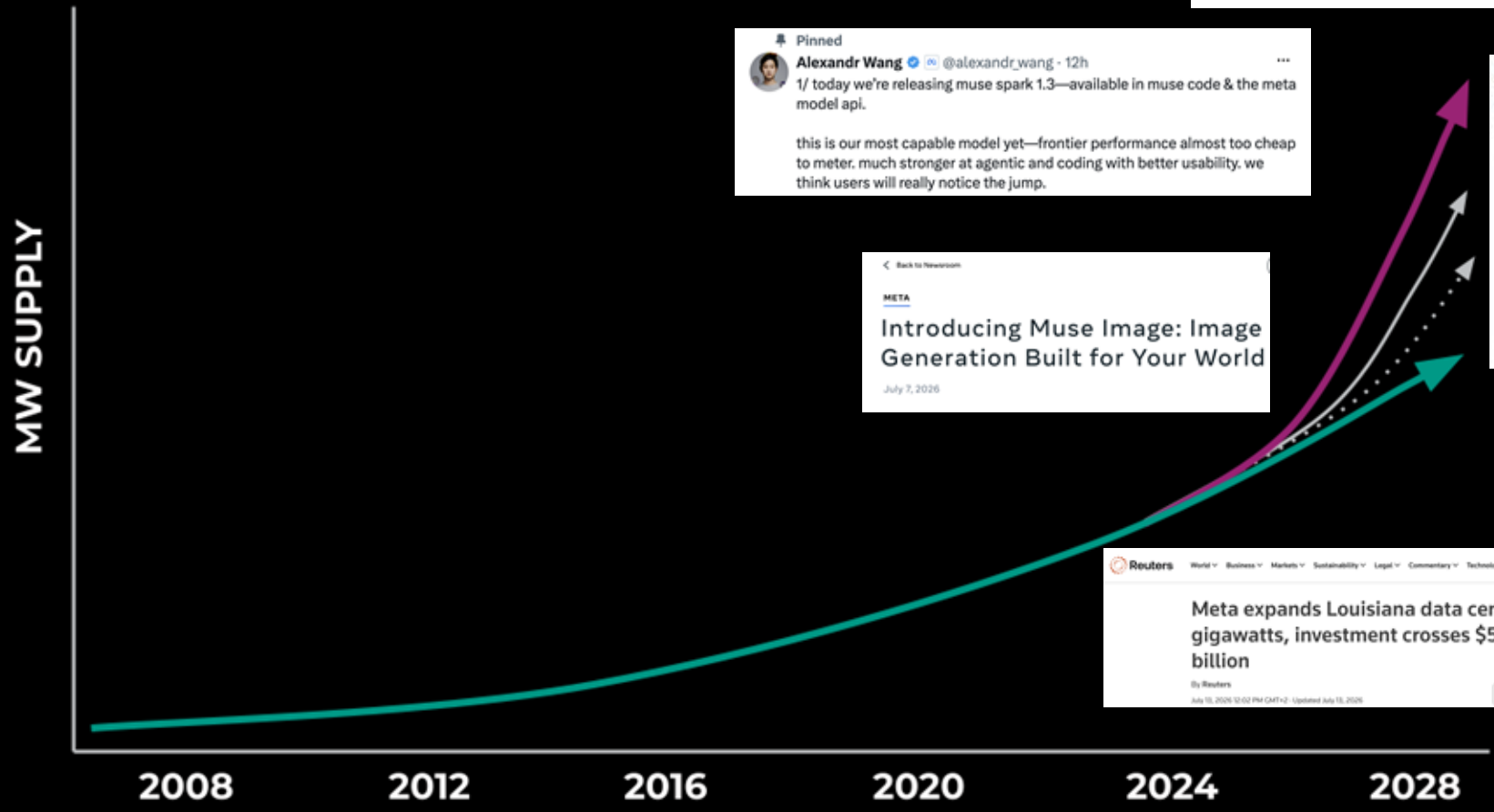
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Source Data: Epoch AI, "AI data centers" (AI data center timelines, accessed Sept. 29, 2026).
*Epoch AI counts operational tents for Meta Prometheus

Meta’s journey to Scale-Across Networking for AI



Compute Supply… Explosion!



META

Introducing Muse: The World's First Personal AI Agent Built for Everyone

September 8, 2026

LISTEN TO ARTICLE

Pinned

Alexandr Wang @alexandr_wang · 12h

1/ today we're releasing muse spark 1.3—available in muse code & the meta model api.

this is our most capable model yet—frontier performance almost too cheap to meter. much stronger at agentic and coding with better usability. we think users will really notice the jump.

META

Introducing Muse Image: Image Generation Built for Your World

July 7, 2026

META

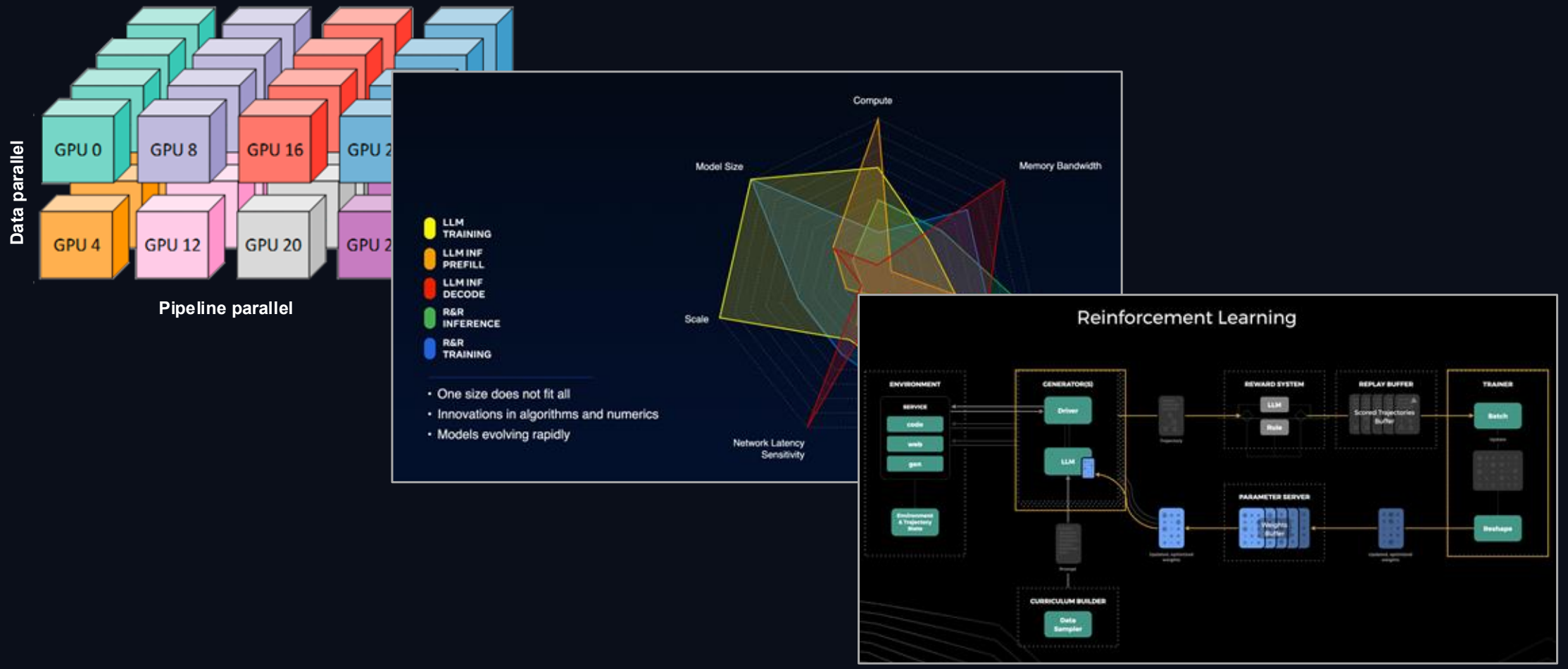
The Future Is for Everyone: Muse for Small Business

Reuters

Meta expands Louisiana data center to 5 gigawatts, investment crosses $50 billion

…. many more DC regions, much faster!

AI Workloads



Every generation on one DC region



A single region now mixes H-shaped halls, single-building designs and tents, alongside third-party cloud capacity.

Meta data centres regions, connected by EBB



Kansas City, MO
Sarpy, NE
Kuna, ID
Prineville, OR
Eagle Mountain, UT
Los Lunas, NM
Mesa, AZ
Fort Worth, TX
Temple, TX

Rosemount, MN
Altoona, IA
DeKalb, IL
Jeffersonville, IN
New Albany, OH
Gallatin, TN
Henrico, VA
Forest City, NC
Stanton Springs, GA
Huntsville, AL

Luleå, Sweden
Odense, Denmark
Clonee, Ireland
Singapore

New construction
Serving traffic

23 Data Center Regions:
19 in NA
3 in Europe
1 in Asia

Rings

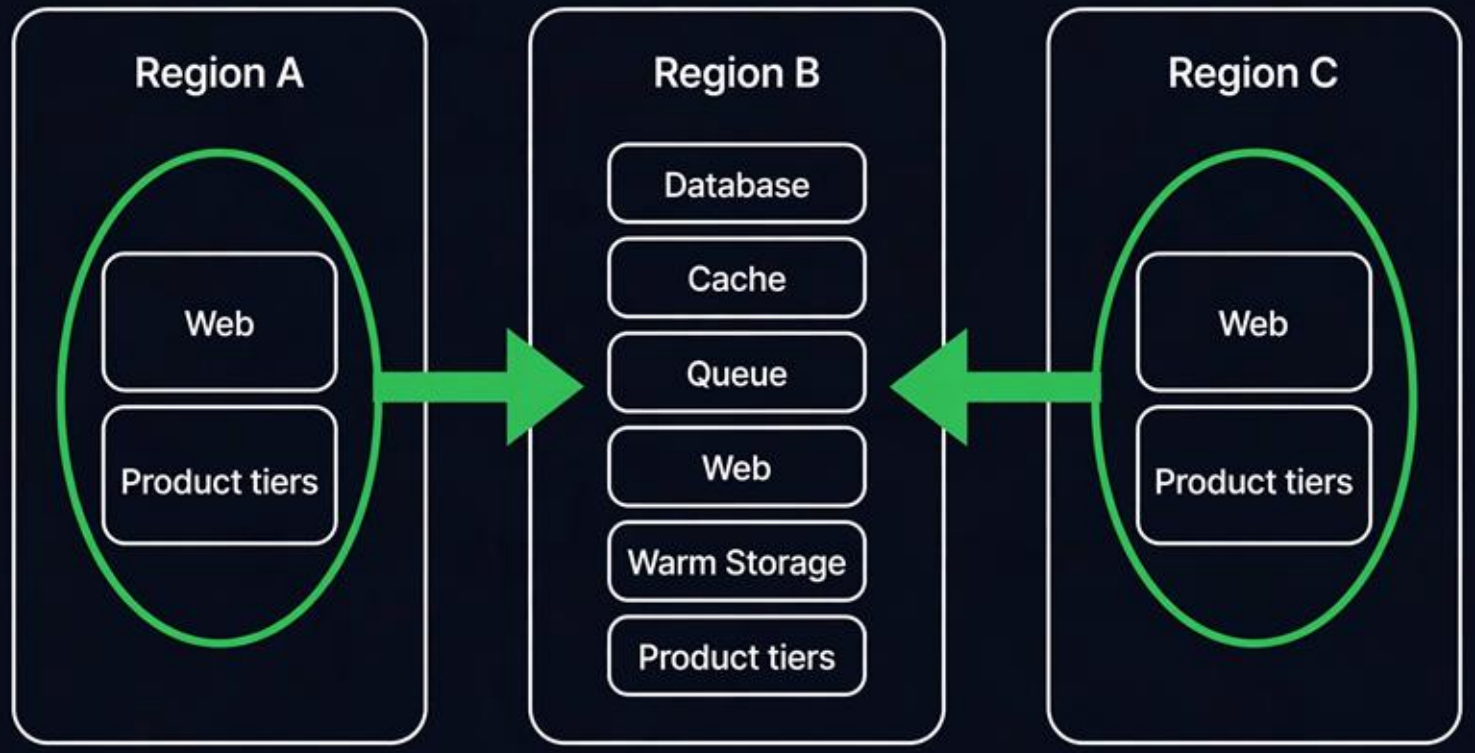


Overlaying fibre topology on the build plan to group regions into “Rings”.

Rings are built in round-trip-time bands: from smaller to higher ms RTT.

Latency follows fibre, not geography.

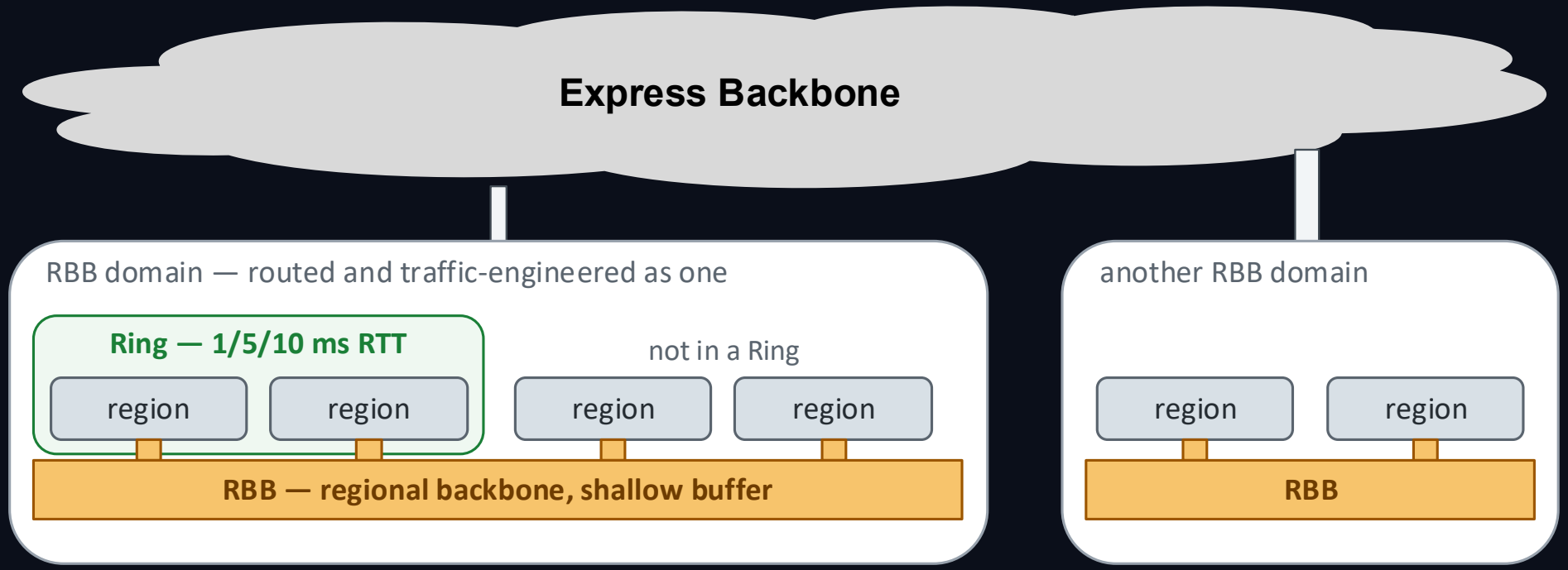
New Idea: One stack per Group of Regions



Neighbouring regions can share a service stack instead of replicating it.

Local reads become Backbone traffic!!

RBB: Regional Backbone blurs DCN/DCI



Same-domain traffic moves from EBB to RBB

Capacity of intra ring is 10x inter ring.

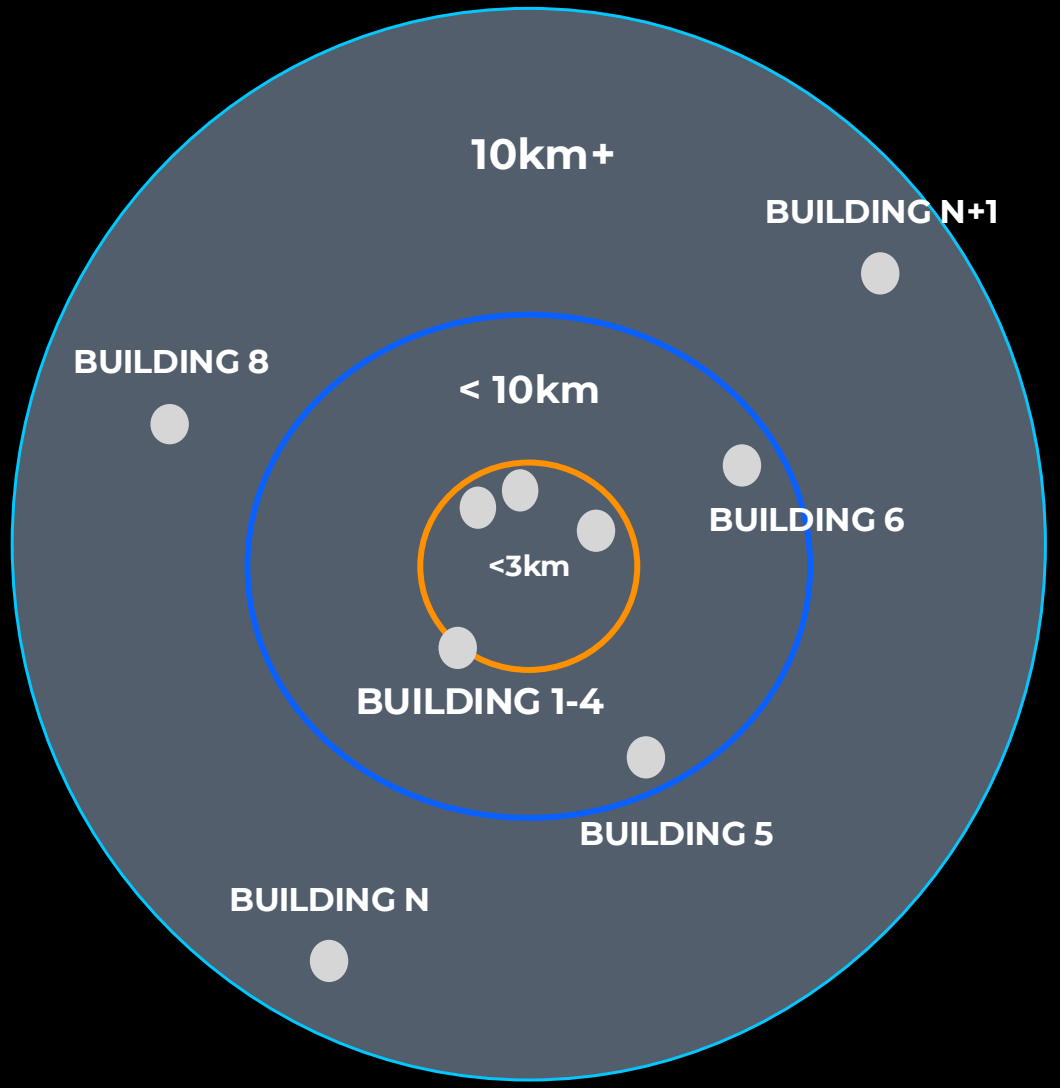
The first trial pair already freed about several MW of baseline power.

Scale-Across: Enabling Large Clusters

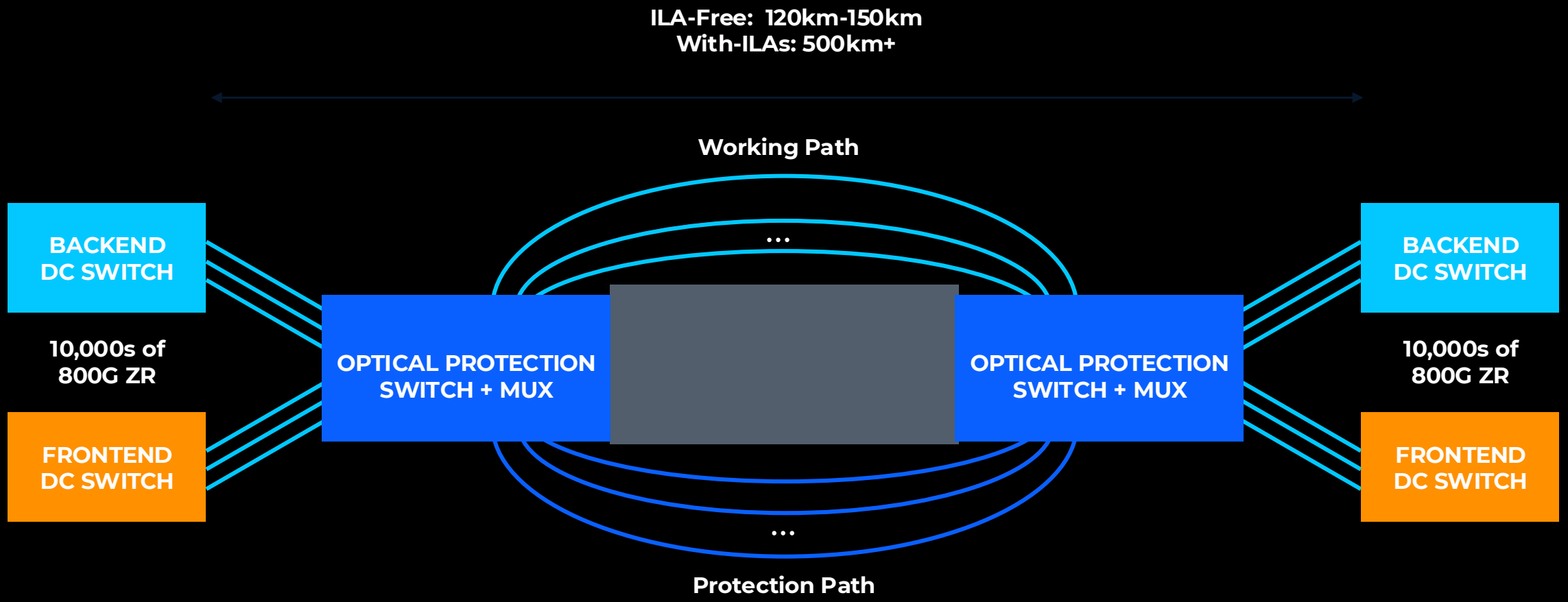
3km reach achievable with FR plugs

10km reach achievable with LR plugs

> 10km requires ZR plugs + DWDM



Connectivity > 10km (Scale-Across or AI-BB)









Open Standards & Adoption

800G-ZR+/1600G ZR+ and OpenROADM/OIF

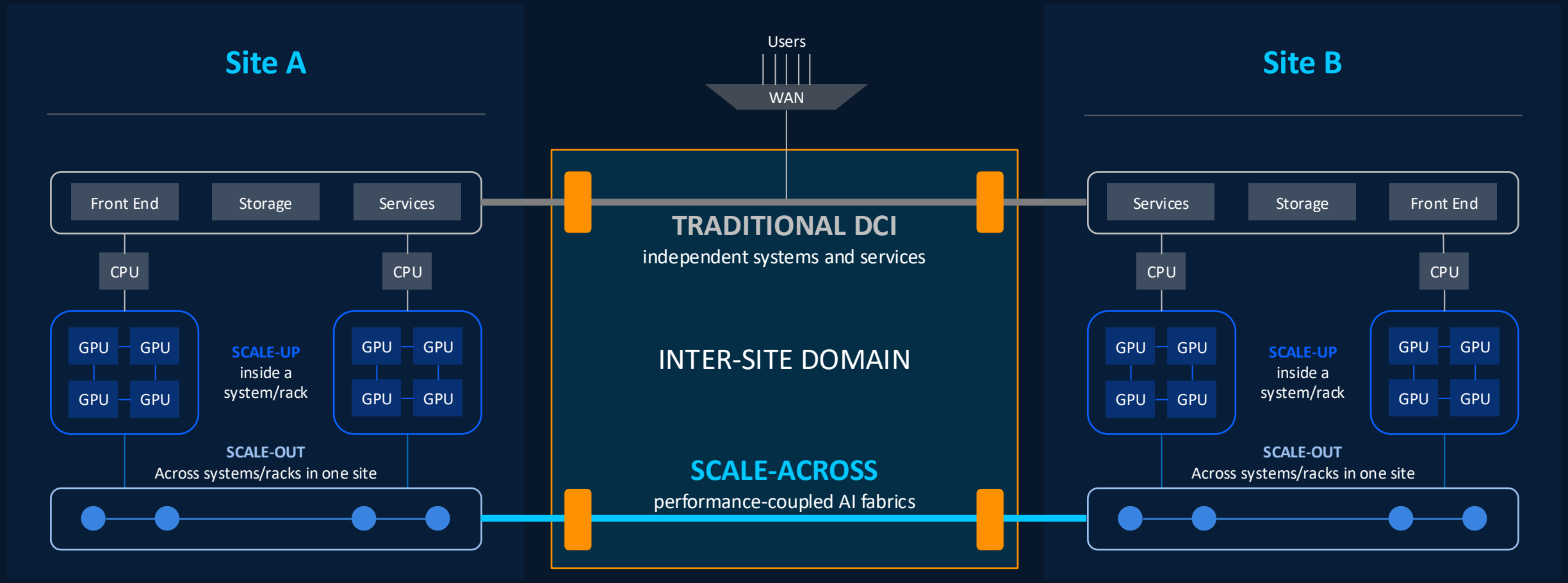
FBOSS and OCP SAI

SRv6 TE and IETF

...

Cisco is an industry innovator, embraces open standards, and a great partner to Meta.

AI networking across systems, fabrics, and sites



Scale-Across puts the inter-site network in the critical path of distributed AI workloads

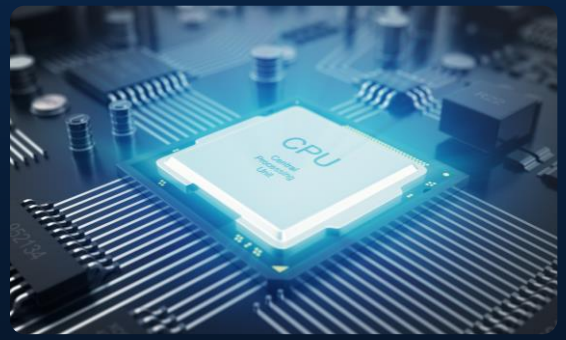
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Different traffic domains require different choices

Traditional DCI

Independent systems

CPU to CPU



Many mixed, bursty, asynchronous flows

Network events affect individual flows or services

Optics technology

Deployment

Scale-across

One coordinated workload

GPU to GPU



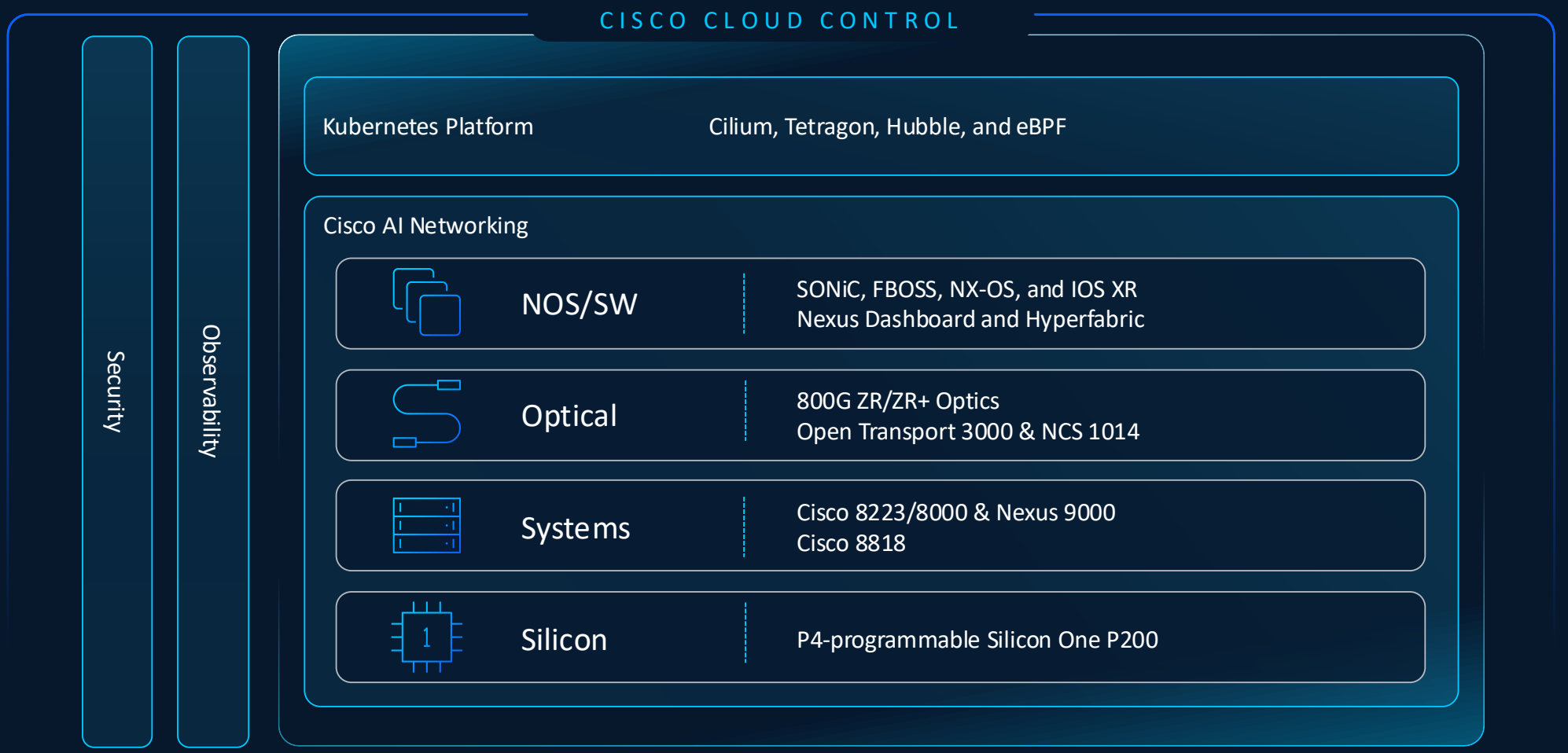
Fewer sustained, synchronized, high-bandwidth flows

Slow and disrupted flows stall progress, idling accelerators

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Cisco delivers the DCI & scale-across stack



Open co-optimized infrastructure at every layer

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A programmable silicon foundation

CISCO CLOUD CONTROL

Cisco AI Networking

NOS/SW	SONiC, FBOSS, NX-OS, and IOS XR Nexus Dashboard and Hyperfabric
Optical	800G ZR/ZR+ Optics Open Transport 3000 & NCS 1014
Systems	Cisco 8223/8000 & Nexus 9000 Cisco 8818
Silicon	P4-programmable Silicon One P200

Software Interfaces
Silicon One SDK with SAI



Cisco Silicon One P200

- Programmability
- Deep Buffering
- Congestion Response
- Security
- Telemetry

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One P200 foundation, form-factor choices

FIXED SYSTEMS

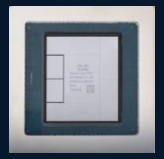
For efficient scale



MODULAR SYSTEMS

For management simplicity





Cisco Silicon One P200

Cisco 8223 and Nexus 9000

51.2T per system

Efficient scale & power
Independent node increments

Consistent unit management
Smaller system-level failure domain

Capacity

Scale

Operations

Cisco 8800 and Nexus 9800

28.8T per line card

Maximum radix

Centralized chassis simplicity
Fewer devices to manage

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Optical transport backbone for DCI and Scale Across

CISCO CLOUD CONTROL

Cisco AI Networking

NOS/SW	SONiC, FBOSS, NX-OS, and IOS XR Nexus Dashboard and Hyperfabric
Optical	800G ZR/ZR+ Optics Open Transport 3000 & NCS 1014
Systems	Cisco 8223/8000 & Nexus 9000 Cisco 8818
Silicon	P4-programmable Silicon One P200

Scalable | **Reliable** | **Power-Efficient**

Pluggable coherent optics (DCOs)



Performance optimized coherent optics



Multi-rail Open Transport System 3000



NCS 1014 multi-haul compact modular system



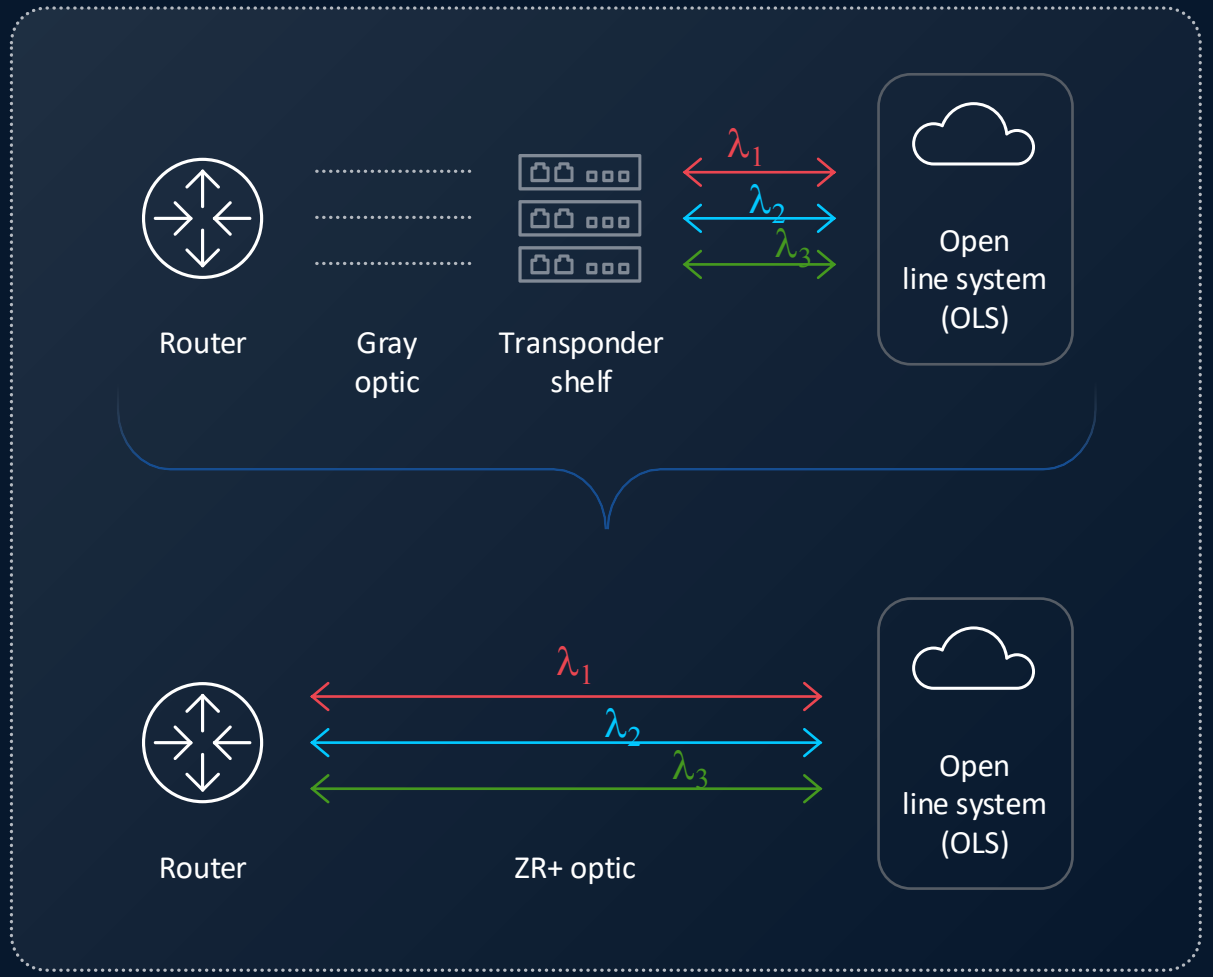
C+L open line system NCS 1010



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Network optimization with coherent pluggable optics





CAPEX reduction

Energy reduction

Space savings

ACG research: TCO savings 40-60% depending on place in network

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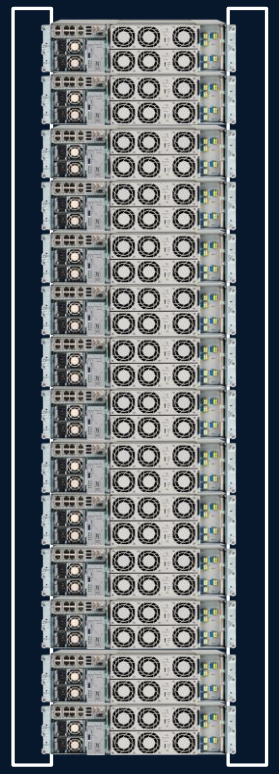
Power-efficient transport scale with Multi-Rail OLS

Single-Rail OLS

1x C-band or L-band Rail per chassis (3RU)

Cisco
NCS 1010

- 350W per C+L Rail
- 7x C+L Rails in a rack
- 32x C+L Rails in 12KW power envelope

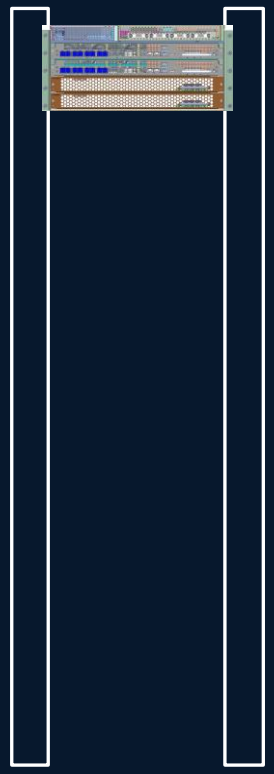


Multi-Rail OLS

4x C+L-band Rails per Line Card

Cisco
Open Transport 3000

- 350W per 4x C+L Rail
- 16x C+L Rails per chassis (5RU)
- 128x C+L Rails per rack, in 12KW power envelope



75%

POWER REDUCTION

80%

SPACE SAVINGS

Massive component density and integration

- Enhanced reliability and scale
- Increased transmission capacity and reach
- Less space, power, cost

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Open and interoperable ecosystems

Driving time to market and economies of scale

Open Coherent Interfaces

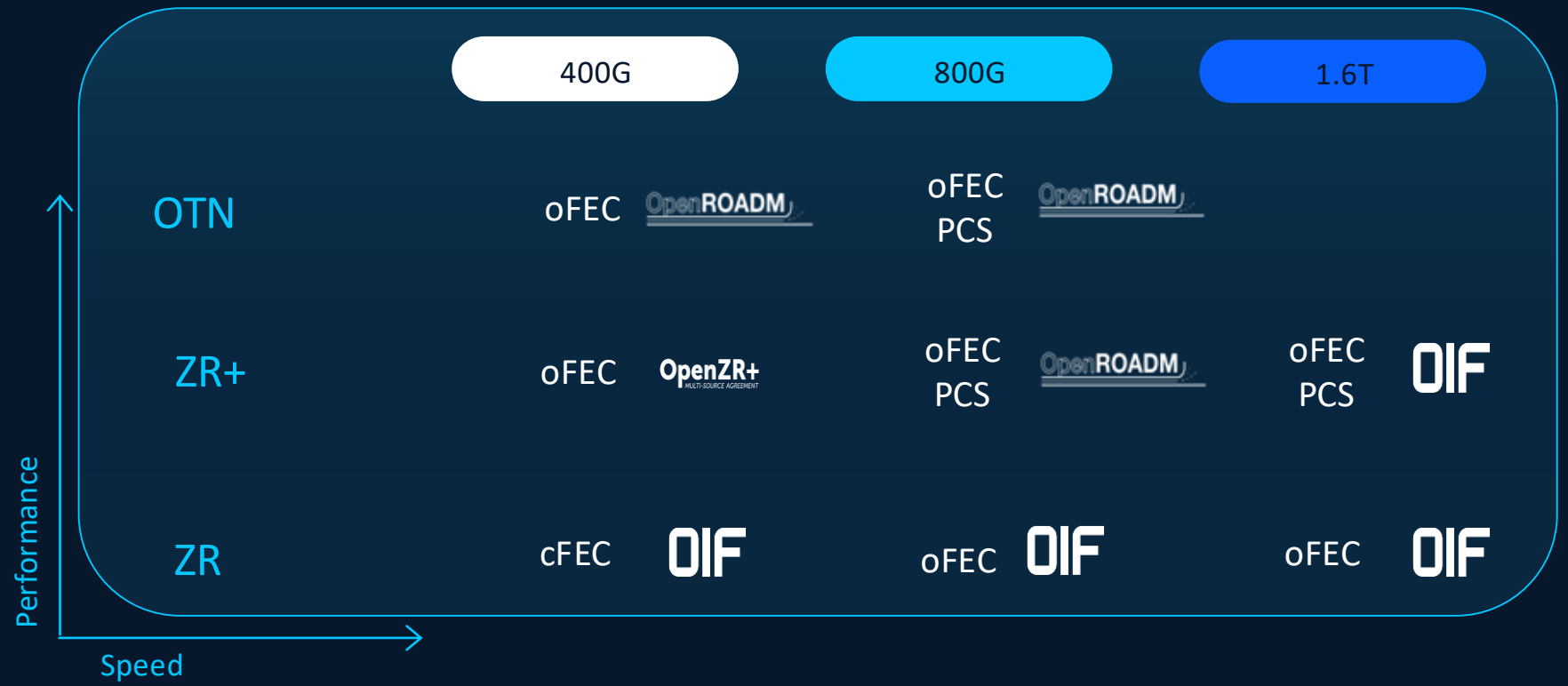
Interoperable standards-based coherent DWDM interfaces supported in MSA standard form factor

Open Management

Vendor-neutral device management;
multi-layer, multi-domain, multi-vendor management

Open Line Systems

Any vendor-coherent transponder or spectrum connects into line system



Cisco led many of these industry efforts and is the leading supplier of pluggable modules

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Maximize scale-across GPU utilization with packet-to-optical co-design



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Extending DCI leadership to distributed AI

AI growth drives DCI and WAN demand

Larger, more distributed data centers increase inter-site and user-facing traffic

Scale-across puts the network in the training path

When a coordinated AI workload spans sites, network events affect job progress

An open packet-to-optical architecture is needed

Deep buffering, congestion control, and coherent optics work together through interoperable interfaces

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Thank you!



Visit our booth

Booth C59

Learn how Cisco is open from silicon to workload.
Built for AI at scale



Attend live demos

- Live Scale-Across Demo
- Live Scale-Out 1.6T Demo with G300 & Cisco Optics
- Open Networking with SONiC



Attend Cisco Sessions

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